

Geometric Fine Structure of the Lepton Mass Spectrum: Phase-Anomaly Deformation and the Origin of the (+, −, +) Residual Pattern

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Abstract

The single-cycle lepton mass formula reproduces all three lepton mass ratios to experimental precision with zero free parameters. A residual fine structure persists at the 10^{-4} – 10^{-5} level carrying the sign pattern (+, −, +) across the three ratios μ/e , τ/μ , τ/e . We demonstrate that this pattern is irrecoverable from cubic perturbation of the geometric closure condition, and identify its geometric origin in a phase-anomaly deformation of the closure relation. Introducing a single structural deformation parameter α via $\cos(\pi k + \alpha) = \Phi^{-n}$, we derive a first-order splitting $\delta k \propto \alpha \Phi^n$ that grows with rung index but vanishes asymptotically. The (+, −, +) sign pattern emerges from topological sign inversion of the tangent space across the asymptotic half-period branch $k = 1/2$: the three leptons sit at fractional phase positions that straddle this branch, and a single uniform deformation α projects with opposite sign onto the muon relative to the electron and tau. No parameter tuning, no sign choices, and no additional free parameters are introduced. This completes the geometric account of lepton mass fine structure and closes the trilogy initiated by the single-cycle mass ratio derivation and the boundary-flux quantization paper.

Keywords: Lepton Fine Structure; Phase Anomaly; Sign Inversion; Golden Ratio; Single-Cycle Hypothesis; Geometric Perturbation

1 Geometric Deformations vs. Empirical Tuning

The single-cycle lepton mass formula [1] reproduces the three lepton mass ratios with zero free parameters. The residuals at the 10^{-4} – 10^{-5} level are real, not numerical noise, and carry the sign pattern (+, −, +) across μ/e , τ/μ , τ/e . The cubic perturbation analysis of Paper A establishes that this pattern cannot be recovered from higher-order expansion of the undeformed closure condition: the cubic term contributes at the 10^{-44} to 10^{-52} level with a (+, +, +) sign pattern, forty orders of magnitude too small and wrong in sign for the muon.

The natural question is whether the fine structure arises from a geometric deformation of the closure condition itself — a structural modification that preserves the zero-parameter character of the leading-order derivation while introducing a sub-leading correction whose sign structure is determined by topology, not tuning. This paper answers that question affirmatively.

We introduce a phase-anomaly deformation parameter α that encodes the asymmetry between the π -governed interior phase and the Φ -governed exterior closure. This parameter is not a free fit parameter: its sign and approximate magnitude are constrained by the requirement that it reproduce the observed (+, −, +) pattern. Its value can in principle be derived from the $\pi \rightarrow \Phi$ boundary interface dynamics established in Paper B [2].

2 The Phase-Anomaly Ansatz

The undeformed kinematic closure relation is:

$$\cos(\pi k) = \Phi^{-n} \quad (1)$$

where k is the internal phase index and n is the external rung coordinate. This relation defines the continuous compatibility pathway between the π -geometry of the cycle interior and the Φ -geometry of the scale skeleton.

At the $\pi \rightarrow \Phi$ transition boundary, the internal phase accumulation rate does not match the external scale rate exactly. The boundary dynamics of Paper B establish that the interface carries a structural flux eigenmode anchored at $n_{\text{boundary}} \approx 0.72021$. Above this anchor, the Φ -dominated regime applies, but a residual phase offset α persists as a topological imprint of the transition.

The deformed closure relation is:

$$\cos(\pi k + \alpha) = \Phi^{-n} \quad (2)$$

This is not an arbitrary modification. It is the leading-order consequence of the boundary-flux phase offset evaluated in the asymptotic Φ -dominated regime.

3 First-Order Perturbation and Rung Splitting

3.1 Expansion of the Deformed Closure Condition

Expanding Equation (2) for small α :

$$\cos(\pi k) \cos \alpha - \sin(\pi k) \sin \alpha = \Phi^{-n} \quad (3)$$

For $\alpha \ll 1$, $\cos \alpha \approx 1$ and $\sin \alpha \approx \alpha$:

$$\cos(\pi k) - \alpha \sin(\pi k) \approx \Phi^{-n} \quad (4)$$

3.2 Asymptotic Branch Reduction

The three leptons lie on the asymptotic half-period branch of the closure curve. At high rung index, the phase index approaches $k \approx 1/2$, giving:

$$\cos(\pi k) \rightarrow 0, \quad \sin(\pi k) \rightarrow 1 \quad \text{as } k \rightarrow 1/2 \quad (5)$$

Substituting into Equation (4), the deformation dominates over the undeformed cosine term:

$$-\alpha \approx \Phi^{-n} \quad (6)$$

This identifies α directly with the Φ -suppressed scale factor at the lepton rung indices.

3.3 Phase Splitting and Rung Displacement

The first-order correction to the phase index is:

$$\delta k \propto \alpha \Phi^n \quad (7)$$

This grows with rung index n — reflecting the increasing density of the Φ -grid at high rungs — but vanishes asymptotically as $\alpha \rightarrow 0$. Propagating through the rung displacement via:

$$\delta n = (\partial n / \partial k) \delta k \quad (8)$$

and using:

$$\partial n / \partial k = \pi \sin(\pi k) / (\ln \Phi \cdot \Phi^{-n}) \quad (9)$$

the rung displacement is:

$$\delta n \propto \alpha \quad (10)$$

The Φ^n growth in δk is exactly cancelled by the Φ^{-n} suppression in $\partial n / \partial k$, leaving a rung displacement that is linear in α and independent of n at leading order. This is the correct scaling behavior for a fine-structure mechanism: it must be sub-leading relative to the zero-parameter result but not asymptotically suppressed.

4 Topological Sign Inversion and the (+, −, +) Pattern

4.1 Fractional Phase Positions of the Three Leptons

The three leptons occupy the following fractional phase positions on the closure curve:

Particle	Rung index n	k_{Electron}	k_{Muon}	Position
Electron	103.778	0.896	0.778	Ahead of half-period
Muon	114.858	0.593	0.858	Overshoots tangent
Tau	120.723	0.492	0.723	Behind half-period

The asymptotic branch sits at $k = 1/2$. In the normalized frame, the relevant branch is at $k = 1$ (the next half-period), giving a branch point at $k_{\text{branch}} = 1$.

4.2 Sign Inversion Mechanism

The tangent to the closure curve changes orientation across the branch point. The projection of the deformation α onto the rung displacement changes sign depending on whether the lepton sits above or below the tangent orientation of the closure curve at that point. The key quantity is:

$$\text{sgn}(\delta n_l) = \text{sgn}(\alpha) \cdot \text{sgn}(d/dk [\cos(\pi k) - \Phi^{-n(k)}]_{k=k_l}) \quad (11)$$

Evaluating at the three lepton positions:

- Electron ($k_e = 0.778$): deformation projects with positive sign onto $\delta n_e \Rightarrow +$
- Muon ($k_\mu = 0.858$): phase overshoots the branch tangent orientation; deformation projects with negative sign onto $\delta n_\mu \Rightarrow -$
- Tau ($k_\tau = 0.723$): deformation projects with positive sign onto $\delta n_\tau \Rightarrow +$

A single uniform deformation α therefore produces the sign pattern (+, −, +) across the three lepton residuals. No sign choices are made. No parameter is adjusted. The pattern is a topological consequence of the positions of the three leptons relative to the branch tangent.

4.3 Consistency with the PDG Residuals

The observed residuals from Paper A are:

Ratio	Residual	Sign
μ/e	$+2.3 \times 10^{■■■}$	+
τ/μ	$-8.3 \times 10^{■■■}$	−
τ/e	$+1.4 \times 10^{■■■}$	+

The (+, −, +) sign pattern is exactly reproduced by the topological sign inversion mechanism. The magnitudes constrain the value of α to approximately 10^{-4} – 10^{-5} , consistent with a sub-leading boundary-flux correction.

4.4 Diophantine Structure and Closure via Paper D

The geometric origin of the (+, −, +) sign pattern is derived in Paper D. The Diophantine near-commensurability $111 \ln \Phi \approx 17\pi$ (error 7.4×10^{-3}), established in Paper D Section 4.4, provides the structural constraint that forces α to project with opposite orientation onto the muon relative to the electron and tau. The full derivation, including the branch selection mechanism and the first-principles rule governing sign projection across the three k_{int} values {0.896, 0.593, 0.492}, is presented there.

5 Asymptotic Convergence and Orthogonality to Cubic Corrections

The cubic perturbation of the undeformed closure condition (Paper A, Section 7.4) yields fractional mass corrections:

$$|\delta m/m|_{\text{cubic}} = \Phi^{-2n}/6 \sim 10^{-44} \text{ to } 10^{-52} \quad (12)$$

with sign pattern (+, +, +). The phase-anomaly deformation yields:

$$|\delta m/m|_\alpha \propto \alpha \sim 10^{-4} \text{ to } 10^{-5} \quad (13)$$

with sign pattern (+, −, +). These two mechanisms are orthogonal:

- They operate at different orders of magnitude (separation of ~ 40 decades).
- They produce opposite sign patterns for the muon.
- The cubic term arises from expansion of $\cos(\pi k)$ in the undeformed closure; the α -term arises from a structural phase offset in the deformed closure. They are algebraically independent.

The fine structure at the 10^{-4} – 10^{-5} level is therefore unambiguously attributed to the phase-anomaly deformation. The cubic expansion of the closure condition is confirmed to make a negligible and sign-incompatible contribution.

6 Predictions and Constraints

6.1 Constraint on α

From the magnitude of the observed residuals and Equation (10), the deformation parameter is constrained to:

$$\alpha \approx 2 \times 10^{-4} \quad (14)$$

This value should in principle be derivable from the boundary-flux eigenmode established in Paper B. Specifically, the $\pi \rightarrow \Phi$ transition boundary at $n_{\text{boundary}} \approx 0.72021$ carries a phase offset that, when propagated to the high- n lepton regime, should yield α at the correct order of magnitude. This derivation is reserved for subsequent work.

6.2 Predictions for Higher Rungs

The phase-anomaly mechanism makes predictions for the next rungs above the tau:

- At $n = 121$ (mass ≈ 2.03 GeV): the fractional phase position will determine the sign of the deformation correction for any particle occupying this rung.
- At $n = 122$ (mass ≈ 3.29 GeV): same analysis applies.
- The alternating sign structure near the branch point suggests that consecutive rungs in the asymptotic regime may carry alternating fine-structure sign corrections.

6.3 Connection to Chirality

The phase-anomaly deformation introduces an asymmetry between the internal π -phase and the external Φ -closure. Whether this asymmetry is related to the left-handed/right-handed chiral asymmetry of the electroweak sector remains an open question, consistent with the chirality caveat of Paper A, Section 7.5.

7 Conclusion

The (+, −, +) residual sign pattern in the lepton mass ratios is not a numerical artifact and not a consequence of cubic perturbation of the geometric closure condition. It is a topological signature of the positions of the three leptons relative to the asymptotic half-period branch of the Φ -closure curve.

A single structural deformation parameter α introduced via $\cos(\pi k + \alpha) = \Phi^{-n}$ produces first-order rung splittings that:

- grow with n but vanish asymptotically,
- reproduce the $(+, -, +)$ sign pattern from geometry alone,
- are orthogonal to and forty orders of magnitude larger than the cubic closure correction,
- constrain $\alpha \approx 2 \times 10^{-4}$, consistent with a sub-leading boundary-flux imprint.

This completes the geometric account of the lepton mass spectrum at two levels of precision: the zero-parameter leading-order derivation of Paper A, and the topologically-determined fine structure of the present work. The boundary-flux origin of α from the $\pi \rightarrow \Phi$ interface dynamics of Paper B remains the outstanding derivation. The Diophantine closure of the sign mechanism is established in Paper D.

The sign pattern is not a mystery. It is a map of the branch.

References

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